

SAMUEL LEVENTHAL

CONTACT INFORMATION

email sl@samleventhal.com
phone 1.801.808.6428
website samleventhal.com
github github.com/sam-lev
scholar [googlescholar.samleventhal.com](https://scholar.google.com/citations?user=sl@samleventhal.com)

ABOUT

Ph.D. in Computer Science from the University of Utah's Scientific Computing and Imaging Institute, working at the intersection of topology, geometry, and machine learning. My research builds mathematical structure directly into learning systems: topological neural operators that encode conservation laws architecturally rather than through penalty terms (NeurIPS 2026, under review), learned persistence filtrations that improve graph neural networks under heterophily, and hierarchical topology-based GNNs for segmentation and representation learning in scientific imaging (ICIP, IEEE TVCG). Applied work spans diagnostic models for lymphoma classification developed with ARUP Laboratories and Mayo Clinic (AJCP) and physics-informed conditional GANs trained at scale within PAVE (IEEE/ACM, SC19), an in-situ framework I co-developed coupling path-traced visualization to distributed PyTorch training on DOE leadership-class machines. The through-line is structure imposed at the level of architecture, which is to say models that are interpretable and physically consistent by construction.

EDUCATION

Ph.D. Computer Science	2024	University of Utah School of Computing & Scientific Computing and Imaging Institute
B.Sc. Physics Minor: Philosophy	2013	University of Utah Department of Physics and Astronomy & College of Humanities
B.Sc. Mathematics Minor: Comp. Sci.	2013	University of Utah Department of Mathematics & School of Computing

DISSERTATION COMMITTEE

Valerio Pascucci (Chair), Christopher R. Johnson, Dave Pugmire, Bei Wang Phillips, Aditya Bhaskara

PUBLICATIONS

PEER REVIEWED CONFERENCE PAPERS

2026 Topological Neural Operators – L. Bastian, S. Leventhal, M. Hajji, T. Birdal. "Topological Neural Operators". NeurIPS, 2026 (under review) arXiv:2606.09806, 2026.

2023 Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks – Leventhal, Samuel, Attila Gyulassy, Valerio Pascucci, and Mark Heimann. "Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks." International Conference on Image Processing. 2023.

2013 A Search for Quantum Like Structuring in Keplerian Systems – Samuel Leventhal and Stephan LeBohec. "A Search For Quantum Like Structuring in Keplerian Systems." University of Utah Undergraduate Research Abstracts Journal, Vol. 13, Book 1. 2013

PEER REVIEWED JOURNAL PAPERS

2022 Exploring Classification of Topological Priors with Machine Learning for Feature Extraction – Leventhal, Samuel, Attila Gyulassy, Mark Heimann, and Valerio Pascucci. "Exploring Classification of Topological Priors with Machine Learning for Feature Extraction." IEEE Transactions on Visualization and Computer Graphics (2023).

2020 Improving Augmented Human Intelligence to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma Cases –Mohlman, Jeffrey S., Samuel D. Leventhal, Taft Hansen, Jessica Kohan, Valerio Pascucci, and Mohamed E. Salama.

"Improving augmented human intelligence to distinguish Burkitt lymphoma from diffuse large B-cell lymphoma cases." American journal of clinical pathology 153, no. 6 (2020): 743-759.

2019 High-throughput feature extraction for measuring attributes of deforming open-cell foams. – Petruzza, Steve, Attila Gyulassy, Samuel Leventhal, John J. Baglino, Michael Czabaj, Ashley D. Spear, and Valerio Pascucci. "High-throughput feature extraction for measuring attributes of deforming open-cell foams." IEEE Transactions on Visualization and Computer Graphics 26, no. 1 (2019): 140-150.

2018 Application of a Convolutional Neural Network to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma – Mohlman, Jeffrey, Samuel Leventhal, Aniketh Venkat, Attila Gyulassy, Valerio Pascucci, and Mohamed Salama. "Application of a Convolutional Neural Network to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma." American Journal of Clinical Pathology 150, no. suppl.1 (2018): Page S119-S119.

2011 Entropy Is Not Our Anomie – Leventhal, Samuel. the Sponge, Volume 2, Issue 1. p5-p11. 2011

PEER REVIEWED WORKSHOP PAPERS

2022 Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks – Leventhal, Samuel, Attila Gyulassy, Valerio Pascucci, and Mark Heimann. "Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks." In NeurIPS 2022 Workshop: New Frontiers in Graph Learning. 2022.

2019 PAVE: An In Situ Framework for Scientific Visualization and Machine Learning Coupling – Leventhal, Samuel, Mark Kim, and David Pugmire. "Pave: An in situ framework for scientific visualization and machine learning coupling." In 2019 IEEE/ACM 5th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD-5), pp. 8-15. IEEE, 2019.

PAPERS IN PREPARATION

2026 Topological Neural Operators – Leventhal, Samuel, Lennart Bastian, Tolga Birdal, and Mustafa Hajij "Topological Neural Operators". extended version in preparation for JMLR.

2026 Homophily-based Filtration Learning For Graph Neural Networks – Leventhal, Samuel, Attila Gyulassy, Mark Heimann, and Valerio Pascucci. In preparation.

EXPERIENCE

2025–
present Postdoctoral Research – Topological Neural Operators

University of San Francisco

Supervisor: Mustafa Hajij

2017–24 Ph.D. Research Scientist

University of Utah, Scientific Computing & Imaging Institute – Ph.D.

Mentor: Valerio Pascucci

2022 Visiting Research Scientist – GETO GNN: Topologically and Geometrically Constructed and Aware Graph Neural Networks

Lawrence Livermore National Laboratory

Mentors: Mark Heimann and Jayaraman J. Thiagarajan

2021 Teaching Mentorship – Programming for Engineers.

Mentor: Valerio Pascucci

2020 Teaching Mentorship – Advanced Algorithms.

Mentor: Aditya Bhaskara

2019 Visiting Research Scientist – PAVE: An In Situ Framework for Scientific Visualization and Machine Learning Coupling

Oak Ridge National Laboratory

Mentor: Mark Kim and David Pugmire

2018 Visiting Research Scientist – High dimensional space filling and sampling design with improved spectral sampling and applications

Monte-Carlo Integration.

Lawrence Livermore National Laboratory

Mentor: Jayaraman J. Thiagarajan

2016 Independent Study – Topological Distortion as a Quality Measure
for Dimension Reduction

University of Utah

Mentor: Bei Wang

2010-12 Independent Study – Search for Quantum like Structuring in Keplerian
Systems.

University of Utah

Mentor: Stephan LeBohec

OTHER EXPERIENCE

2023 Conference Presentation

IEEE International Conference on Image Processing Graph Signal Processing and
Machine Learning for Interpretable and robust image processing II

2019 Conference Presentation

Super Computing Conference 2019, 5th International Workshop on Data Analysis
and Reduction for Big Scientific Data (DRBSD-5)